
When Does a Small Model Know to Hand Off? Reading Competence from Full-Duplex Speech Models for Real-Time Escalation

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Abstract

1 Full-duplex speech models answer in real time with a small on-device backbone,
2 yet many queries exceed its competence. We ask whether the model’s own hid-
3 den state signals, *before the first output token*, that a query is beyond it — reli-
4 ably enough to hand off to a large cloud model without breaking the live session.
5 Late-layer competence readouts turn brittle after duplex fine-tuning: final-layer
6 probes collapse under distribution shift on text input (leave-one-pool-out math
7 AUC .93→.37), consistent with late-layer specialization for duplex turn control,
8 while verbalized self-assessment collapses on audio input. A mid-network linear
9 probe survives both: leave-one-pool-out math .93, text-calibrated speech deploy-
10 ment .86, and an end-of-turn read in the streaming loop that reaches AUC .887
11 in ~30 ms — before decoding starts. Run live end-to-end in a turn-based loop,
12 the gated system climbs from .422 to .633 heard accuracy at 57% escalation, with
13 the remaining gap dominated by the transcription uplink rather than the gate. The
14 gate then transfers *frozen* to six public speech benchmarks using only label-free
15 threshold quantiles, lifting Speech TriviaQA to .860 live against the model’s offi-
16 cial offline .755. A competence signal sufficient for real-time escalation is already
17 present mid-network in duplex-trained speech models; the standard readouts sim-
18 ply miss it.

19 1 Introduction

20 A full-duplex speech model listens and speaks in the same forward pass, under a conversational
21 deadline of a few hundred milliseconds [Stivers et al., 2009]. That deadline forces a small on-device
22 backbone — and a small backbone routinely faces queries beyond its competence: long-tail facts,
23 expert knowledge, multi-step reasoning. The natural architecture is selective escalation to a large
24 cloud model, but the duplex setting imposes two constraints that existing routing work does not
25 meet: the decision must come *before the first output token* (once the model starts speaking, a wrong
26 answer is already on the air), and the session — audio context, voice, cache — must stay with the
27 local model, which will speak the expert’s answer itself.

28 This paper asks whether the small model already contains the signal this decision needs. We study
29 MiniCPM-o 4.5 [Cui et al., 2026], a 9B omni/duplex fine-tune of Qwen3-8B, against matched-
30 pair controls spanning raw backbones, streaming-omni, audio-only, and frozen-adaptor fine-tunes,
31 and evaluate every standard zero-shot competence readout — decode-time entropy, verbalized self-
32 assessment ($p(\text{True})$), and linear probes on hidden states — with judged failure labels on a frozen

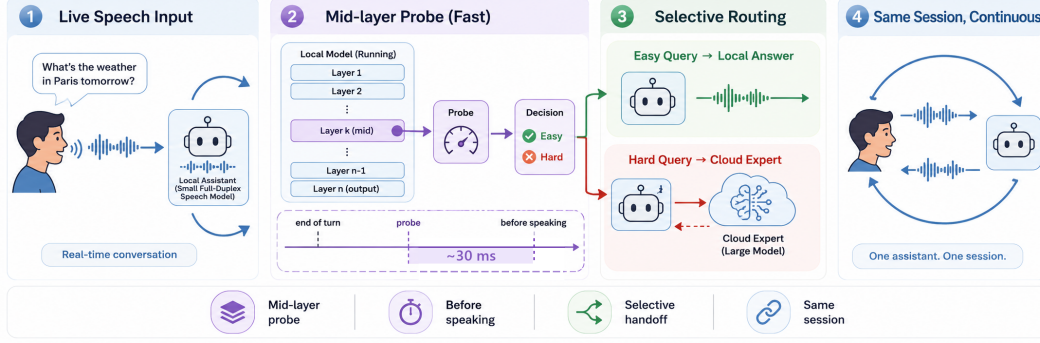


Figure 1: (1) Speech streams live into a small duplex-trained local model. (2) A mid-network (L22) linear read decides in ~ 30 ms — before the first output token — so the expert call is already in flight while local decoding would just be starting. (3) Easy queries stay local; hard ones escalate to a cloud expert. (4) The *same talker speaks* the expert’s answer in one continuous session, improving live spoken sessions (.422 $\rightarrow$.633) and, frozen, public speech benchmarks (Speech TriviaQA .664 $\rightarrow$.860).

pool of public-benchmark queries, in matched text and speech conditions. Throughout, *duplex* names the training regime, not the interaction protocol: we study duplex-trained weights driven through a *turn-based* loop whose only decision point is an end-of-turn read (§6); a deployed talker-on replay leaves the gate unchanged.

Everything runs on a frozen checkpoint — no fine-tuning; the only fitted component is a logistic regression on stored hidden states, calibrated once on text. That minimalism is the point: competing signals read the wrong place (late layers), arrive too late (post-answer verification), or must be trained in (think tokens); the mid-layer read survives all three. Three findings, in increasing order of system-level consequence:

1. **Representation: duplex fine-tuning makes the late-layer readouts brittle; a mid-network signal survives.** The conventional final-layer probe looks strong in distribution (AUC .82) but is mostly a query-type shortcut: under leave-one-pool-out transfer it *inverts* on held-out math (.93 $\rightarrow$.37). The damage is specific to the late-layer last-token position on text input — absent in raw backbones and non-duplex fine-tunes — consistent with late-layer specialization for duplex turn control. Mid-network (L22 of 36) the same linear read transfers at .93, and the mid-network optimum generalizes to a second, architecturally disjoint duplex family (hybrid-Mamba NemotronLabs-VoiceChat-11B).
2. **Modality: the mid-layer signal transfers text $\rightarrow$ speech in end-to-end models.** A probe calibrated on cheap text data reads speech-input hidden states at .86; the control that isolates the mechanism is Freeze-Omni, an adapter on frozen identical weights, where cross-modal transfer is dead ($\leq .60$). Verbalized self-assessment shows the mirror-image failure — it collapses on audio input — so the mid-layer probe is the one readout that survives both modalities. The transfer holds on real human recordings, not just TTS.
3. **System: the signal is available before the first token and works live, in and out of distribution.** The L22 read costs 20 ms (text) / 45 ms (audio) against time-to-first-token of 36/68 ms; in the streaming loop an end-of-turn read reaches AUC .887 in ~ 30 ms. Run live end-to-end — expert consulted mid-conversation, answer injected, talker speaks it — heard accuracy climbs .422 $\rightarrow$.633 at 57% escalation, and the frozen gate transfers to six public speech benchmarks with label-free thresholds, lifting Speech TriviaQA to .860 against the model’s official offline .755.

Extended analyses — readout-damage controls, router baselines, escalation-channel diagnostics, latency mechanics — are in the appendix.

65 2 Related Work

66 **Full-duplex real-time dialogue.** End-to-end spoken dialogue spans native duplex designs —
 67 dGSLM [Nguyen et al., 2023], Moshi [Défossez et al., 2024], GLM-4-Voice [Zeng et al., 2024],
 68 single-LLM native full-duplex [Fang et al., 2026] — streaming omni models [Xu et al., 2025a, Cui
 69 et al., 2026], frozen-backbone adapters [Wang et al., 2024a], and audio fine-tunes [Chu et al., 2024].
 70 A parallel line adds explicit turn-taking control: FlexDuo [Liao et al., 2025], MinMo [Chen et al.,
 71 2025], and Freeze-Omni’s state heads decide *when the local model should speak*, at 120–250 ms
 72 cadences that show a live loop tolerates far more decision latency than our ~ 30 ms read [cf. Riera
 73 et al., 2026, Wu et al., 2026]. Our gate decides the orthogonal question — *whether the local model*
 74 *should be the one answering* — and the head-based controllers’ shared choice of a last-layer readout
 75 is exactly the point our results show duplex fine-tuning damages.

76 **Does the model know what it knows?** $p(\text{True})$ self-evaluation [Kadavath et al., 2022], verbalized
 77 uncertainty [Lin et al., 2022], and truthfulness probes on hidden states [Azaria and Mitchell, 2023,
 78 Burns et al., 2023, Marks and Tegmark, 2024] established that correctness is readable internally;
 79 intermediate layers carry more of it than the output distribution [Orgad et al., 2025]. Semantic
 80 entropy [Kuhn et al., 2023, Farquhar et al., 2024] needs multiple sampled generations; Semantic
 81 Entropy Probes [Kossen et al., 2024] amortize it into a hidden-state probe — the closest signal to
 82 ours, but read post-hoc on text-only half-duplex models. Multimodal uncertainty fusion [Zhang
 83 et al., 2026] asks a downstream question; we ask *where* in a duplex model a modality-robust read
 84 survives at all.

85 **Selective escalation and routing.** FrugalGPT [Chen et al., 2023], HybridLLM [Ding et al., 2024],
 86 RouteLLM [Ong et al., 2024], and confidence-based cascades [Xue et al., 2025, Hu et al., 2024, Li
 87 et al., 2026] route from query-side features or preference data; AutoMix [Aggarwal et al., 2024]
 88 escalates by *post-answer* self-verification, which a voice deadline forbids. Both assumptions —
 89 query-side features, and the large model answering the user directly — fail in full-duplex voice,
 90 where the session lives in the small model (query-side ceiling quantified in §4.1). A second family
 91 trains the decision *into the weights* as control tokens: pause and thought tokens [Goyal et al., 2024,
 92 Zelikman et al., 2024], fused think/no-think modes [Yang et al., 2025, DeepSeek-AI et al., 2025],
 93 and the built-in think gate MiniCPM-o 4.5 itself ships [Cui et al., 2026]. These decide *more compute*,
 94 not *which model*, require fine-tuning the talker, and bake the policy into the checkpoint — changing
 95 budget or expert means retraining. Our gate is the pluggable complement: a logistic read over a
 96 frozen checkpoint, refit from stored activations in minutes, transferring across modality and model
 97 family (§7), and it beats the model’s own trained think gate at matched budget (Table 2). Learning-
 98 to-defer [Madras et al., 2018] and early exit [Schuster et al., 2022] decide inside one model.

99 **The gap.** Duplex work decides when to speak; routing work decides which model — from the
 100 query surface, after a first answer, or via tokens trained into the weights; uncertainty work asks
 101 whether the model is confident, almost always offline and in text. No prior work we know of asks
 102 whether a full-duplex speech model *already* exposes a pre-answer, modality-robust, internal compe-
 103 tence signal early enough to drive a live small-to-large handoff while preserving the session.

104 3 Experimental Setup

105 **Target model and matched-pair controls.** The system target is **MiniCPM-o 4.5** (9B; omni +
 106 full-duplex fine-tune of Qwen3-8B), bf16, greedy decoding, seed 42. To attribute effects to the *kind*
 107 of fine-tune rather than the backbone, we assemble ten models spanning raw backbones, a second du-
 108 plex generation (MiniCPM-o 2.6), a streaming-omni fine-tune (Qwen2.5-Omni-7B), an audio-only
 109 fine-tune (Qwen2-Audio-7B), and a frozen-backbone adapter model (Freeze-Omni, LLM weights
 110 byte-identical to its raw control); the full roster is Table 4 (appendix). The comparison statistic is
 111 always the within-pair difference.

Table 1: Signals on MiniCPM-o 4.5, judged failure labels. In-dist = calibration-split AUC (probes 5-fold out-of-fold); LOPO math = leave-one-pool-out AUC on held-out math; audio = the same readout under speech input (final/mid-layer probe: audio-input LOPO math / text-only calibration read on audio states). Full per-signal detail: Appendix B.

signal	in-dist	LOPO math	audio input
max entropy@4	.696	$\approx$ chance on raw backbones	—
$p(\text{True})$ -pre	.807	trap .945 (probe: .232)	collapses
$p(\text{True})$ -post (full draft)	.899	—	degrades
final-layer probe	.822	.372 (inverts)	.936 (intact)
mid-layer (L22) probe	.866	.931	.855 (text-calibrated)

Query pool, labels, and audio arm. 600 queries in five pools, frozen before any gate experiment (360 calibration / 240 test): *trap* = SimpleQA [Wei et al., 2024] (the target fails 100%), *hard-knowledge* = MMLU-Pro [Wang et al., 2024b], *easy-fact* = TriviaQA [Joshi et al., 2017], *easy-chat* = Dolly/Alpaca [Conover et al., 2023, Taori et al., 2023], *hard-math* = GSM8K tail + MATH-500 [Cobbe et al., 2021, Hendrycks et al., 2021, Lightman et al., 2024]; public benchmarks only, no LLM-generated queries. Adequacy labels come from an LLM judge [Zheng et al., 2023] with structured outputs, blind to the gate and validated by a stronger re-judge (agreement .962 text / .945 audio; every headline number moves $\leq .010$ under the stronger labels); the escalation expert is gpt-5.5 at low reasoning effort. The audio arm renders the same frozen pool to speech (single-voice TTS, content unchanged — only modality varies), following Spoken-SQuAD / VoiceBench practice [Li et al., 2018, Chen et al., 2024]; audio-side findings are additionally validated on real human recordings (SD-QA, Faisal et al., 2021) and on the external benchmarks’ official audio (§7). Judge prompts, pins, and spend: Appendix L–M.

Gate signals (no model training). (i) decode scalars: max token entropy / margin over the first k decode steps; (ii) linear probes: logistic regression on a single hidden state, position $\times$ layer swept, fit on the calibration split only; (iii) verbalized self-evaluation [Kadavath et al., 2022]: $p(\text{True})$ -pre asks “would you answer this correctly?” before answering, $p(\text{True})$ -post asks about a drafted answer; $P(\text{Yes})$ is read from the first token’s distribution.

4 Reading Competence from Duplex Models

4.1 In distribution every readout looks fine; transfer reveals the shortcut

Table 1 compares the candidate signals on MiniCPM-o 4.5. In distribution the final-layer probe reaches .822 — but that number is mostly a *query-type* shortcut: the same hidden state classifies a query’s source pool at 95.8%, a pool-identity oracle alone reaches AUC .715, and appending true pool labels to the probe adds nothing (.822 $\rightarrow$.821). The decisive test is leave-one-pool-out (LOPO) transfer, which we treat as the primary metric throughout: held-out math *inverts* to .372, and the mean escalation score on a 100%-fail trap pool is .232. An external query-surface router makes the same point from the other side: under identical labels a TF-IDF router reaches only OOF AUC .669 (below the pool oracle), collapses to .38–.57 under LOPO, and 100 $\times$ the data on a public routing benchmark buys back only .04 of AUC (Appendix D). Instance-level competence information lives in the model’s internal state, not on the query surface.

Verbalized self-assessment transfers differently: *pre-answer* $p(\text{True})$ catches the trap pool the probe misses (escalation score .945 vs. the probe’s .232) — but timing is essential: asked *after* drafting, the model endorses its own confident-wrong answer (.360). Post-draft $p(\text{True})$ is the strongest single signal (.899) but requires paying for a full draft, which a voice deadline cannot afford.

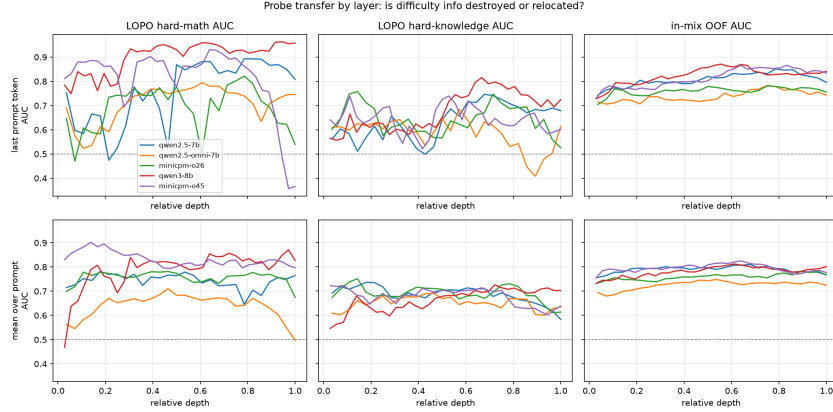


Figure 2: The core representational result: LOPO math AUC by layer and position. On the duplex model the standard (final-layer, last-token) read inverts on text input while mid-network layers hold $\geq .9$; raw backbones show no cliff. The deployed gate reads L22.

146 4.2 The damage is duplex-specific; the signal survives mid-network

147 Matched pairs attribute the probe failure to the *kind* of fine-tune, not the backbone: the raw Qwen
148 backbone, label-matched to the duplex model’s fail rates, holds LOPO math at .825 where the duplex
149 model sits at .372; streaming-omni and audio-only fine-tunes are undamaged; a second duplex gen-
150 eration (MiniCPM-o 2.6) reproduces the collapse (Appendix B). A layer $\times$ position sweep localizes
151 it (Figure 2): the collapse is specific to (late layer $\times$ last token) on text input — mean-pooled reads
152 at the same depth are fine, and mid-network the last-token read recovers to **.931 at L22**, near the raw
153 backbone. This is the position a streaming head must re-purpose to encode turn-control state, and
154 the pattern is consistent with late-layer specialization for duplex turn control overwriting the read-
155 out while the underlying information survives. The mid-network optimum is not a MiniCPM quirk:
156 on NemotronLabs-VoiceChat-11B — a hybrid-Mamba duplex model, architecturally disjoint — the
157 same recipe peaks mid-band at L30–34 of 56 and transfers to the external pools at AUC .70–.79 with
158 a quarter of the calibration data (Appendix C).

159 4.3 Text-calibrated, speech-deployed — in end-to-end models only

160 Probes trained on text hidden states and scored on speech-input hidden states locate where the
161 representation becomes modality-shared: early layers are modality-specific (.54–.60), and from
162 $\sim 50\%$ depth text $\rightarrow$ audio transfer plateaus at .82–.87 (.855 at the deployed L22). This replicates
163 on MiniCPM-o 2.6 and cross-family on Qwen2.5-Omni. The decisive control is **Freeze-Omni**:
164 identical frozen LLM weights with audio attached by an adapter — and cross-modal transfer is dead
165 (.52–.60). The modality-shared mid-layer core is *created by training the backbone on the modality*;
166 the cheap recipe “calibrate on text, deploy on speech” is valid for end-to-end models and invalid
167 for adapter-attached ones. Verbalized introspection shows the mirror-image failure: on audio input,
168 pre-answer $p(\text{True})$ collapses on both MiniCPM duplex generations while both non-duplex controls
169 stay intact; controlled arms show the verbal self-check runs over text-token pathways that audio em-
170 beddings do not feed — duplicating the question as text restores it — while a linear probe reads the
171 same information off the same forward pass at .93 (Appendix B). All audio-side findings replicate
172 on 200 real human recordings (SD-QA), ruling out TTS artifacts. **The mid-layer probe is the one**
173 **readout that survives every training regime that can converse**; the live system deploys the same
174 L22 read plus two zero-latency pooled positions (the *deployed* probe; Appendix B).

Table 2: Deployment policies, frozen test split ($n=240$); latencies measured.

policy	acc	lat. mean (s)	P50	P95
fast-only	.588	3.0	3.5	4.2
all-THINK (built-in)	.637	22.4	17.2	60.1
gated-think @.33	.613	12.1	3.6	47.6
gated-cloud @.15	.688	5.3	3.6	10.1
gated-cloud @.33	.787	6.5	3.6	20.8
gated-cloud @.50	.858	7.0	3.6	24.4

5 From Signal to Gate

The decision exists before decoding starts. The gate is one 4096-d dot product on a hidden state the forward pass already computes. Measured on H100, the L22 decision lands in **20 ms (text) / 45 ms (audio)** against time-to-first-token of 36/68 ms: $T_{\text{gate}} < T_{\text{first token}}$, far inside the 200–300 ms turn-taking budget [Stivers et al., 2009], and the cloud call can launch while the remaining $\sim 40\%$ of prefill and the whole decode still run ($p(\text{True})$ -pre costs an extra prefill, $p(\text{True})$ -post a full draft: 1.9/3.5 s; per-layer cost curves in Appendix E).

Accuracy–cost tradeoff. On the frozen test split ($n=240$): small-only .588, big-only (gpt-5.5) .917. Gated escalation beats random escalation at every operating point (Figure 6, appendix); the balanced tier at a 33% budget reaches **.779–.787**, recovering $\sim 58\%$ of the small→big gap at one third of the big-only cost. In distribution the gate is statistically on par with a pool-oracle (type-only) router — as the type-shortcut audit predicts — so its case over type routing rests on transfer, where the type shortcut collapses (§4.1) and the mid-layer signal does not.

Against the model’s own thinking mode. MiniCPM-o 4.5 ships a built-in reasoning mode — self-escalation in place. Gated-cloud escalation dominates all-THINK on both accuracy and latency at every escalation rate (gated-cloud@.33: .787 at mean 6.5 s vs. all-THINK .637 at 22.4 s; Table 2). The reason is mechanistic: the gate predominantly flags knowledge and trap failures, which extra compute cannot fix — thinking helps math and hurts chat — so external escalation is necessary, not merely better.

6 Live End-of-Turn Escalation

The streaming loop. The protocol is turn-based: audio streams in 1 s chunks, but the talker speaks only after the user’s turn ends and the model’s native listen/speak head is not engaged. Per-chunk probe scores are strictly weaker than the offline read (best chunk statistic AUC .764) and miss short utterances whose critical entity lands in the final chunk. The fix is an **end-of-turn read**: after the last audio chunk, prefill an assistant turn containing a single space and read L22 at the assistant-start position. This read reaches **AUC .887** — above the offline reference (.843) — in ~ 30 ms, still before the first output token, and is the loop’s only decision point; escalation-rate thresholds are quantiles of this single score. Speculative mid-stream prefetch was measured and dropped (an expert answer to a partial transcript answers the full question only .07–.51 of the time; Appendix F). On escalation the talker speaks a stall phrase, its *own* transcription of the audio goes to the expert (gpt-5.5, reasoning effort fixed low — higher effort buys zero accuracy on the escalated population at double the latency), and the returned answer is injected as context for the talker to relay. On the deployed channel the relay is clean: model wrong, correct answer injected, comply = 1.00 ($n=24$); more broadly the competence signal also predicts relay compliance (comply .42→.75 across gate-score quartiles) — a model that knows it doesn’t know is willing to listen. Conflicting-injection and framing controls: Appendix F.

The live result. We ran 240 frozen test queries $\times$ four escalation tiers, each a complete turn-based spoken session, scored on *heard accuracy* — what a user would actually have heard. Headline

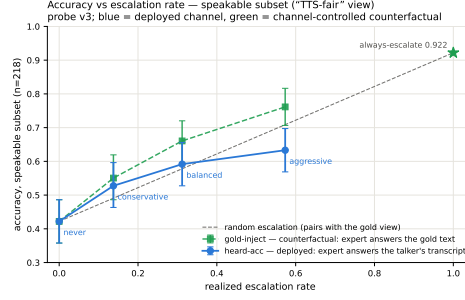


Figure 3: Live dual-view tradeoff ($n=218$ speakable subset). Blue: honest heard accuracy of the live system (paired-bootstrap CIs). Green: the channel-controlled counterfactual (same sessions, escalated rows re-scored with gold-text expert answers) with its paired random reference (dashed). The view-to-view gap is the speech-channel price; the floor-to-floor gap to offline operation is the audio + streaming-answer tax.

numbers use a pre-registered input-side filter (21 formula-LaTeX ids + 1 broken recording removed; spoken formulas are destroyed by any TTS render before the system hears them). Heard accuracy climbs monotonically, $.422 \rightarrow .528 \rightarrow .592 \rightarrow .633$ at 0/14/31/57% escalation, every tier’s gain significant (paired bootstrap; Figure 3). The remaining gap to the channel-controlled counterfactual (escalated rows re-scored with gold-text expert answers) is dominated by *transcription-uplink* errors, not gate failure: the channel costs $+.023/+0.069/+0.128$ across tiers, and in the channel-controlled view the gate clears its paired random reference at every arm. Live arm accuracies on this pool carry a replication-noise floor of $\pm .02-.03$, so no claim here rests on a sub-3-point live delta. Full tables (both views, both pools), latency profiles, uplink diagnostics, and two boundary query classes (false-premise, real-time) are in Appendix F. A talker-on replay (237/239 turns barge-in-seeded) leaves the frozen probe intact: AUC $.871/.856$ vs. $.866$ headless (Appendix G). A TTS-on re-run of the balanced arm puts first *audible* audio at p50 .55 s / p99 .64 s after end of turn on local answers and 22–29 ms on escalated ones — the canned stall starts speaking while the expert call is in flight (Appendix F).

7 Frozen Transfer to Public Speech Benchmarks

The out-of-distribution test the type-shortcut audit demands: we take the deployed gate *frozen* — weights, features, and layer choice fit once on the internal calibration mix — and run the complete live loop of §6 on six public speech benchmarks: Speech TriviaQA, Speech Web Questions, Llama Questions, and Reasoning QA from OpenAudioBench [Li et al., 2025] (spoken renderings of Joshi et al., 2017, Berant et al., 2013, and the set of Nachmani et al., 2024), SD-QA (real dialectal human speech) [Faisal et al., 2021], and VoiceBench AlpacaEval [Chen et al., 2024, Taori et al., 2023]. All audio is the benchmarks’ official pre-rendered audio; no query ever touches probe calibration. The only per-pool adaptation is *label-free*: tier thresholds are quantiles of the probe’s own score distribution on each pool, consuming no labels and no judge calls. It buys budget control, not discrimination: one absolute threshold frozen on the internal pool still beats matched-rate random on four of five pools, but fires at 2–48% across them (Appendix H). Where an official protocol exists we score with it (the OpenAudioBench judge; VoiceBench’s 1–5 judge), and our chat-mode control reproduces the official anchors on-scale.

The frozen gate improves all five QA benchmarks (Table 3): thresholds fire at sane rates, every curve rises monotonically, and 50% escalation lifts the floor $+.11-.29$. On Speech TriviaQA the 9B system reaches **.860 live** — against its own official offline $.755$ and $.629$ for Qwen3-Omni-30B [Xu et al., 2025b] at $3.3\times$ the parameters: selective cloud escalation outperforms larger standalone backbones. AUC transfers at $.68-.81$ against $.879$ internally, and the cleanest evidence that the signal is difficulty rather than surface is cross-lingual: an English-only calibration expansion moved *Chinese* Reasoning QA from AUC $.621$ to $.683$, the largest per-pool gain. On Llama Questions —

Table 3: Main transfer results: the frozen gate on six public speech benchmarks. Left: live heard accuracy at realized escalation 0–50% ($\pm .02$ – $.03$ floor); ceiling = GPT-5.5 on gold text; official = the model’s reported offline score. Right: gold-inject view, matched 30% budget — random / gate / benefit oracle (Appendix H). Judges in parentheses.

pool	n	official	live heard, by tier					ceiling	AUC	gold-inject @30%		
			0%	15%	30%	50%	rand			gate	oracle	
TriviaQA (OAB)	250	.755	.664	.732	.800	.860	.968	.789	.755	.836	.964	
Web Questions (OAB)	250	.702	.57	.628	.680	.732	.864	.785	.657	.700	.868	
Llama Q. (OAB)	250	—	.84	.904	.904	.948	.928	.806	.864	.912	.948	
SD-QA (ours)	200	—	.51	.600	.720	.795	.930	.792	.639	.745	.815	
Reasoning zh (ours)	202	—	.59	.653	.713	.772	.871	.683	.674	.728	.891	
AlpacaEval (VB 1–5)	199	4.8	3.99	3.96	4.23	4.35	4.96	—	—	—	—	

single-type, so no shortcut to exploit — **selective escalation matches the expert at half the expert calls**: the probe’s split is decisive (kept-local .976 vs. escalated .696, $z=6.46$), the expert’s whole advantage lands where the probe sends it (.696→.888, $p<.0001$), and always-escalate buys nothing measurable over selective; matched-rate precision reaches 95% of the base-rate cap (Appendix I).

Two honest negatives. **AlpacaEval**: the gate does not beat random — open-ended instruction following has no “missing fact” event for a retrieval-failure detector to read. **Web Questions and SD-QA are flat** under the deployed-probe refit: on Web Questions strict alias-based labels decouple “wrong” from “uncertain” (even the gold-text ceiling is .864), and the failures the gate ranks highest are ones the expert misses too (expert-fixable share .66 at the 15% cut vs. .75 pool-wide). The external channel cost is small (heard-vs-gold gap .035–.056 at aggressive vs. .175 internally), and SD-QA’s escalated subset converts at .82–.84 on real dialectal speech. Routing can also *reduce* tail latency, replacing slow wrong local decodes with short expert round-trips (Reasoning QA P90 13.3→10.9 s; Appendix I).

8 Discussion

What the signal detects. The pre-answer probe is specifically a *retrieval-failure* detector, strongest on missing-knowledge and long-tail-trap failures where the empty lookup is an observable event in the model’s state. Execution failures (getting stuck mid-derivation) surface in *decode-time* signals instead, and metacognitive failures — questions with false premises, where no failure event occurs in the model’s experience — are invisible to every signal we tested, query-surface routers included (fire rate .18 vs. trap’s .90; Appendix F). This scope motivates a decode-time check behind the pre-answer gate.

Limitations. (i) One phrasing per $p(\text{True})$ variant. (ii) Real-speech validation covers scripted read speech in one dialect. (iii) Formula-symbolic content is out of scope for the audio arm — destroyed by any TTS render, an input-side limit distinct from the transcription-uplink loss. (iv) The second-family replication covers mid-band readability and frozen-recipe transfer, not the full matched-pair attribution or the live loop. (v) Labels come from one judge family, bounded by a stronger re-judge ($\text{deltas} \leq .010$), without human labels. (vi) The live sweep is one session per query per tier ($\pm .02$ – $.03$ replication floor); the talker-on replay (Appendix G) leaves concurrent-with-generation reads open.

9 Conclusion

Duplex-trained speech models carry a competence signal their standard readouts miss: a mid-network linear read that survives the readout damage, transfers from text calibration to speech, and lands tens of milliseconds before the first output token. Deployed live, it turns a 9B talker into a system that knows when to hand off; frozen, it lifts public speech benchmarks past models three times its size. Reading the right layer makes real-time escalation possible.

References

- Pranjal Aggarwal, Aman Madaan, Ankit Anand, Srividya Pranavi Potharaju, Swaroop Mishra, Pei Zhou, Aditya Gupta, Dheeraj Rajagopal, Karthik Kappaganthu, Yiming Yang, Shyam Upadhyay, Manaal Faruqui, and Mausam. AutoMix: Automatically mixing language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2310.12963.
- Amos Azaria and Tom Mitchell. The internal state of an LLM knows when it’s lying. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023. arXiv:2304.13734.
- Jonathan Berant, Andrew Chou, Roy Frostig, and Percy Liang. Semantic parsing on Freebase from question-answer pairs. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2013. WebQuestions.
- Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. Discovering latent knowledge in language models without supervision. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2212.03827.
- Lingjiao Chen, Matei Zaharia, and James Zou. FrugalGPT: How to use large language models while reducing cost and improving performance. *arXiv preprint arXiv:2305.05176*, 2023.
- Qian Chen, Yafeng Chen, Yanni Chen, et al. MinMo: A multimodal large language model for seamless voice interaction. *arXiv preprint arXiv:2501.06282*, 2025.
- Yiming Chen, Xianghu Yue, Chen Zhang, Xiaoxue Gao, Robby T. Tan, and Haizhou Li. VoiceBench: Benchmarking LLM-based voice assistants. *arXiv preprint arXiv:2410.17196*, 2024.
- Yunfei Chu, Jin Xu, Qian Yang, Haojie Wei, Xipin Wei, Zhifang Guo, et al. Qwen2-Audio technical report. *arXiv preprint arXiv:2407.10759*, 2024.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021. GSM8K.
- Mike Conover, Matt Hayes, Ankit Mathur, Jianwei Xie, Jun Wan, Sam Shah, Ali Ghodsi, Patrick Wendell, Matei Zaharia, and Reynold Xin. Free dolly: Introducing the world’s first truly open instruction-tuned LLM. Databricks blog, 2023. databricks-dolly-15k.
- Junbo Cui, Bokai Xu, Chongyi Wang, Tianyu Yu, Weiyue Sun, Yingjing Xu, et al. MiniCPM-o 4.5: Towards real-time full-duplex omni-modal interaction. *arXiv preprint arXiv:2604.27393*, 2026.
- DeepSeek-AI, Daya Guo, et al. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, and Neil Zeghidour. Moshi: a speech-text foundation model for real-time dialogue. *arXiv preprint arXiv:2410.00037*, 2024.
- Dujian Ding, Ankur Mallick, Chi Wang, Robert Sim, Subhabrata Mukherjee, Victor Rühle, Laks V. S. Lakshmanan, and Ahmed Hassan Awadallah. Hybrid LLM: Cost-efficient and quality-aware query routing. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2404.14618.
- Fahim Faisal, Sharlina Keshava, Md Mahfuz ibn Alam, and Antonios Anastasopoulos. SD-QA: Spoken dialectal question answering for the real world. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, 2021. arXiv:2109.12072.
- Qingkai Fang, Shoutao Guo, and Yang Feng. BayLing-Duplex: Native full-duplex speech dialogue with a single autoregressive LLM. *arXiv preprint arXiv:2606.14528*, 2026.
- Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630:625–630, 2024.

326 Sachin Goyal, Ziwei Ji, Ankit Singh Rawat, Aditya Krishna Menon, Sanjiv Kumar, and Vaishnavh
327 Nagarajan. Think before you speak: Training language models with pause tokens. In *ICLR*, 2024.

328 Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song,
329 and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. In
330 *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks*, 2021.
331 arXiv:2103.03874.

332 Qitian Jason Hu, Jacob Bieker, Xiuyu Li, Nan Jiang, Benjamin Keigwin, Gaurav Ranganath, Kurt
333 Keutzer, and Shriyash Kaustubh Upadhyay. RouterBench: A benchmark for multi-LLM routing
334 systems. *arXiv preprint arXiv:2403.12031*, 2024.

335 Mandar Joshi, Eunsol Choi, Daniel S. Weld, and Luke Zettlemoyer. TriviaQA: A large scale dis-
336 tantly supervised challenge dataset for reading comprehension. In *Proceedings of the 55th Annual
337 Meeting of the Association for Computational Linguistics (ACL)*, 2017.

338 Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez,
339 Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language mod-
340 els (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

341 Jannik Kossen, Jiatong Han, Muhammed Razzak, Lisa Schut, Shreshth Malik, and Yarin Gal. Se-
342 mantic entropy probes: Robust and cheap hallucination detection in LLMs. *arXiv preprint
343 arXiv:2406.15927*, 2024.

344 Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for
345 uncertainty estimation in natural language generation. In *International Conference on Learning
346 Representations (ICLR)*, 2023. arXiv:2302.09664.

347 Chia-Hsuan Li, Szu-Lin Wu, Chi-Liang Liu, and Hung-yi Lee. Spoken SQuAD: A study of miti-
348 gating the impact of speech recognition errors on listening comprehension. In *Interspeech*, 2018.
349 arXiv:1804.00320.

350 Hao Li, Yiqun Zhang, Zhaoyan Guo, Chenxu Wang, Shengji Tang, Qiaosheng Zhang, Yang Chen,
351 Biqing Qi, Peng Ye, Lei Bai, Zhen Wang, and Shuyue Hu. LLMRouterBench: A massive bench-
352 mark and unified framework for LLM routing. *arXiv preprint arXiv:2601.07206*, 2026.

353 Tianpeng Li, Jun Liu, Tao Zhang, Yuanbo Fang, Da Pan, Mingrui Wang, et al. Baichuan-Audio:
354 A unified framework for end-to-end speech interaction. *arXiv preprint arXiv:2502.17239*, 2025.
355 Introduces OpenAudioBench.

356 Borui Liao, Yulong Xu, Jiao Ou, Kaiyuan Yang, Weihua Jian, Pengfei Wan, and Di Zhang. Flex-
357 Duo: A pluggable system for enabling full-duplex capabilities in speech dialogue systems. *arXiv
358 preprint arXiv:2502.13472*, 2025.

359 Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan
360 Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. In *International
361 Conference on Learning Representations (ICLR)*, 2024. arXiv:2305.20050; source of the MATH-
362 500 subset.

363 Stephanie Lin, Jacob Hilton, and Owain Evans. Teaching models to express their uncertainty in
364 words. *Transactions on Machine Learning Research*, 2022. arXiv:2205.14334.

365 David Madras, Toniann Pitassi, and Richard Zemel. Predict responsibly: Improving fairness and
366 accuracy by learning to defer. In *Advances in Neural Information Processing Systems (NeurIPS)*,
367 2018.

368 Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language
369 model representations of true/false datasets. In *Conference on Language Modeling (COLM)*, 2024.
370 arXiv:2310.06824.

371 Eliya Nachmani, Alon Levkovitch, Roy Hirsch, Julian Salazar, Chulayuth Asawaroengchai, Soroosh
372 Mariooryad, Ehud Rivlin, RJ Skerry-Ryan, and Michelle Tadmor Ramanovich. Spoken question
373 answering and speech continuation using spectrogram-powered LLM. In *International Confer-*
374 *ence on Learning Representations (ICLR)*, 2024. arXiv:2305.15255; Spectron, source of the
375 Llama Questions dataset.

376 Tu Anh Nguyen, Eugene Kharitonov, Jade Copet, Yossi Adi, Wei-Ning Hsu, Ali Elkahky, Paden
377 Tomasello, Robin Algayres, Benoît Sagot, Abdelrahman Mohamed, and Emmanuel Dupoux. Gen-
378 erative spoken dialogue language modeling. *Transactions of the Association for Computational*
379 *Linguistics*, 2023. arXiv:2203.16502.

380 Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez,
381 M. Waleed Kadous, and Ion Stoica. RouteLLM: Learning to route LLMs with preference data.
382 *arXiv preprint arXiv:2406.18665*, 2024.

383 Hadas Orgad, Michael Toker, Zorik Gekhman, Roi Reichart, Idan Szpektor, and Yonatan Belinkov.
384 LLMs know more than they show: On the intrinsic representation of LLM hallucinations. In
385 *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2410.02707. TODO
386 verify author list.

387 Pablo Riera, Pablo Brusco, Cristina Kuo, Marcelo Sancinetti, and S. R. K. Branavan. Synchroniza-
388 tion and turn-taking in full-duplex speech dialogue models. *arXiv preprint arXiv:2605.20356*,
389 2026.

390 Tal Schuster, Adam Fisch, Jai Gupta, Mostafa Dehghani, Dara Bahri, Vinh Q. Tran, Yi Tay, and
391 Donald Metzler. Confident adaptive language modeling. In *Advances in Neural Information*
392 *Processing Systems (NeurIPS)*, 2022. arXiv:2207.07061.

393 Tanya Stivers, N. J. Enfield, Penelope Brown, Christina Englert, Makoto Hayashi, Trine Heinemann,
394 Gertie Hoymann, Federico Rossano, Jan Peter de Ruiter, Kyung-Eun Yoon, and Stephen C. Levin-
395 son. Universals and cultural variation in turn-taking in conversation. *Proceedings of the National*
396 *Academy of Sciences*, 106(26):10587–10592, 2009.

397 Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy
398 Liang, and Tatsunori B. Hashimoto. Stanford alpaca: An instruction-following LLaMA model.
399 GitHub repository, 2023. zh split: shibing624/alpaca-zh.

400 Tu Vu, Mohit Iyyer, Xuezhi Wang, Noah Constant, Jerry Wei, Jason Wei, Chris Tar, Yun-Hsuan
401 Sung, Denny Zhou, Quoc Le, and Thang Luong. Freshllms: Refreshing large language models
402 with search engine augmentation. *arXiv preprint arXiv:2310.03214*, 2023.

403 Xiong Wang, Yangze Li, Chaoyou Fu, Lei Xie, Ke Li, Xing Sun, and Long Ma. Freeze-Omni:
404 A smart and low latency speech-to-speech dialogue model with frozen LLM. *arXiv preprint*
405 *arXiv:2411.00774*, 2024a.

406 Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, et al. MMLU-Pro: A more robust and
407 challenging multi-task language understanding benchmark. In *Advances in Neural Information*
408 *Processing Systems (NeurIPS), Datasets and Benchmarks*, 2024b. arXiv:2406.01574.

409 Jason Wei, Nguyen Karina, Hyung Won Chung, Yunxin Joy Jiao, Spencer Papay, Amelia Glaese,
410 John Schulman, and William Fedus. Measuring short-form factuality in large language models.
411 *arXiv preprint arXiv:2411.04368*, 2024. SimpleQA.

412 Donghang Wu, Haoyang Zhang, Chen Chen, Tianyu Zhang, Fei Tian, Xuerui Yang, Gang Yu, Hexin
413 Liu, Nana Hou, Yuchen Hu, and Eng Siong Chng. Chronological thinking in full-duplex spoken
414 dialogue language models. In *Proceedings of the 27th Annual Meeting of the Special Interest*
415 *Group on Discourse and Dialogue (SIGDIAL)*, 2026. arXiv:2510.05150.

416 Jin Xu, Zhifang Guo, Jinzheng He, Hangrui Hu, Ting He, Shuai Bai, Keqin Chen, et al. Qwen2.5-
417 Omni technical report. *arXiv preprint arXiv:2503.20215*, 2025a.

418 Jin Xu, Zhifang Guo, Hangrui Hu, Yunfei Chu, Xiong Wang, et al. Qwen3-Omni technical report.
419 *arXiv preprint arXiv:2509.17765*, 2025b.

420 Leyang Xue, Yao Fu, Luo Mai, and Mahesh K. Marina. HybridServe: Efficient serving of large AI
421 models with confidence-based cascade routing. *arXiv preprint arXiv:2505.12566*, 2025.

422 An Yang et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.

423 Eric Zelikman, Georges Harik, Yijia Shao, Varuna Jayasiri, Nick Haber, and Noah D. Goodman.
424 Quiet-STaR: Language models can teach themselves to think before speaking. *arXiv preprint*
425 *arXiv:2403.09629*, 2024.

426 Aohan Zeng, Zhengxiao Du, Mingdao Liu, Kedong Wang, Shengmin Jiang, Lei Zhao, Yuxiao Dong,
427 and Jie Tang. GLM-4-Voice: Towards intelligent and human-like end-to-end spoken chatbot.
428 *arXiv preprint arXiv:2412.02612*, 2024.

429 Dengjia Zhang, Alexander Martin, William Jurayj, Kenton Murray, Benjamin Van Durme, and Reno
430 Kriz. Unified multimodal uncertain inference. *arXiv preprint arXiv:2604.08701*, 2026.

431 Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang,
432 Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica.
433 Judging LLM-as-a-judge with MT-Bench and chatbot arena. In *Advances in Neural Information*
434 *Processing Systems (NeurIPS), Datasets and Benchmarks*, 2023. arXiv:2306.05685.

435 A Model roster

Table 4: Model roster. “Pair” groups a fine-tune with its raw backbone control; the comparison statistic is always the within-pair difference.

model	type	pair	role
MiniCPM-o 4.5	omni + duplex FT	Qwen3-8B	system target
Qwen3-8B	raw	—	pair control
MiniCPM-o 2.6	duplex FT	Qwen2.5-7B	duplex replication
Qwen2.5-Omni-7B	streaming omni FT	Qwen2.5-7B	omni control
Qwen2.5-7B	raw	—	pair control
Qwen2-Audio-7B	audio FT	Qwen2-7B	audio-FT control
Freeze-Omni	frozen LLM + adapter	Qwen2-7B	adapter control
Qwen2-7B	raw	—	pair control
Mistral-7B-v0.3	raw	—	backbone diversity
GLM4-9B-chat	raw	—	backbone diversity

436 B Extended signal and readout analyses

437 **ROC curves.** Figure 4 shows the calibration-split ROC for the signals of Table 1.

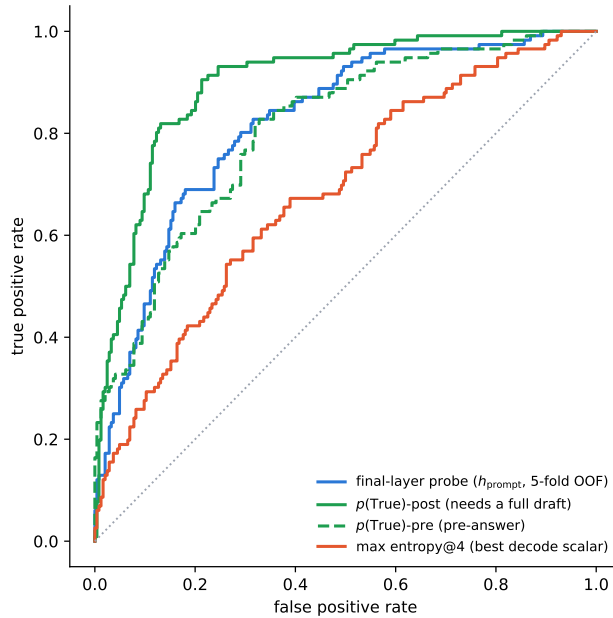


Figure 4: ROC on the calibration split. $p(\text{True})$ -post leads but needs a full drafted answer; the probes and $p(\text{True})$ -pre are pre-answer.

438 **$p(\text{True})$ backbone dependence.** Across backbones, post-draft $p(\text{True})$ beats the trained final-
 439 layer probe on four of five raw/duplex models tested; the exception (GLM4-9B-chat, .685 vs. probe
 440 .798) shows the pattern depends on the backbone’s self-evaluation quality.

441 **Matched pairs.** Table 5: with label coverage matched by subsampling the raw control to the
 442 duplex model’s per-pool fail rates, the raw backbone keeps LOPO math at .825/.809 while duplex
 443 models sit at chance — label coverage is ruled out; the representation changed. Degradation is
 444 ordered raw > omni-streaming > duplex. Table 6 gives the by-depth summary behind Figure 2: the
 445 within-pair deficit at the best mid-layer is only $-.03$ vs. $-.59$ at the final layer; mean-pooled probes

survive to the final layer on duplex models; duplex damage is severe but local, omni-streaming damage mild but diffuse.

Table 5: LOPO transfer (AUC, held-out pool) of the final-layer last-token probe, Qwen2.5-7B family, label-coverage matched.

LOPO pool	Qwen2.5-7B (raw)	Qwen2.5-Omni	MiniCPM-o 2.6	[o4.5 vs Qwen3]
hard-math	0.809	0.745	0.538 $\approx$ chance	0.372 (inverts)
hard-knowledge	0.680	0.613	0.526 $\approx$ chance	0.61

Table 6: LOPO math transfer by depth (last-token probes).

model	best mid-layer	final layer	shape
Qwen3-8B (raw)	0.964 (L33/36)	0.958	plateau, no cliff
MiniCPM-o 4.5 (duplex)	0.931 (L22/36)	0.366	cliff, inverts
Qwen2.5-7B (raw)	0.893 (L21/28)	0.809	mild dip
Qwen2.5-Omni (streaming)	0.794 (L16/28)	0.746	diffuse depression
MiniCPM-o 2.6 (duplex)	0.822 (L21/28)	0.540	cliff

Modality specificity of the cliff. Under audio input the final layer does *not* invert (L35 audio LOPO math .936 vs. .366 text); a speak-mode template control on text input leaves the cliff unchanged (.362) — the operative variable is the modality of the input context, not the output mode.

Audio collapse of verbalized introspection. Table 7. Three controlled arms pin the mechanism on MiniCPM-o 4.5: *not perception* (ASR audit WER .074; $p(\text{True})$ on the model’s own transcript snaps back to .074; the well-heard subset still collapses); *not a context prior* (irrelevant filler audio does not inflate p_{yes}); *binding* (duplicating the question as text alongside the audio fully restores introspection, .034). The verbal instance-check runs over text-token pathways that audio embeddings do not feed, while a linear probe reads the same information off the same forward pass at .93+. The failure implies its own cheap fix, confirmed twice: re-present the query as text before asking (“repeat-then-judge”).

Table 7: Pre-answer introspection under audio input. Trap p_{yes} = mean stated probability of answering correctly on a 100%-fail pool (lower is better).

model	FT type	trap p_{yes} : text $\rightarrow$ audio	audio $p(\text{True})$ -pre AUC
MiniCPM-o 4.5	duplex	.055 $\rightarrow$.556 (collapse)	.786 (trap dead)
MiniCPM-o 2.6	duplex	.196 $\rightarrow$.345	.491 $\approx$ chance
Qwen2.5-Omni	omni-streaming	.279 $\rightarrow$.213 (intact)	.727
Freeze-Omni	frozen + adapter	.165 $\rightarrow$.112 (intact)	.612 (capability caveat)

Real-speech validation. The matched-pair design rerun on 200 real human recordings (VoiceBench SD-QA, USA split) replicates all three audio-side findings: modality tax (.400 $\rightarrow$.450), audio overconfidence (paired p_{yes} shift +.089; failure subset .415 $\rightarrow$.581), and the layer structure (text-calibrated read on real-speech audio holds .76–.80 at L22–L26, peak .800 at L25 — transfer across both content and modality).

Synthesis. End-to-end multimodal training *builds* the shared mid-layer core; duplex-style training additionally *damages* the readouts; omni-streaming gets both right; an adapter alone gets neither (probe readout text-fragile, verbal readout audio-fragile; in both cases knowledge survives and a readout breaks, and in both cases the breakage is duplex-fine-tune-specific).

The deployed probe. The mechanistic probe is a single-position L22 read. The deployed probe adds two zero-latency pooled positions (an 8-token tail mean and the running user-audio mean),

refit on a calibration mix widened to 2,310 public-benchmark queries disjoint from every evaluation benchmark; a pre-registered guard held (frozen internal test AUC .860→.879) and feature selection never saw the externals. In the mid-layer band the in-mix tradeoff optimum is broad (L20–L30) and the mid-layer probe’s in-distribution area advantage is modest — its real advantage is LOPO robustness, which an in-mix test cannot show.

C Second duplex family: NemotronLabs-VoiceChat-11B

The pre-registered prediction test: with the identical recipe (same audio, same judge, 600-query calibration), the layer sweep on this hybrid-Mamba duplex model (56 blocks: 27 Mamba-2 / 4 attention) peaks mid-band at L30–34 (eot-last AUC .714 vs. .693/.682 at the endpoints); stacking the same three positions lifts OOF .714→.761→.790; the frozen-methodology probe transfers to the external pools at AUC .781/.793/.754/.701 — the deployed-probe band reached with a quarter of the calibration data. Boundary notes: the backbone is much weaker at knowledge retrieval (striviaqa local .32 vs. MiniCPM .62, same judge) and English-only (the cross-lingual pool has no analog). The live-loop port and the trained-checkpoint ablation remain future work.

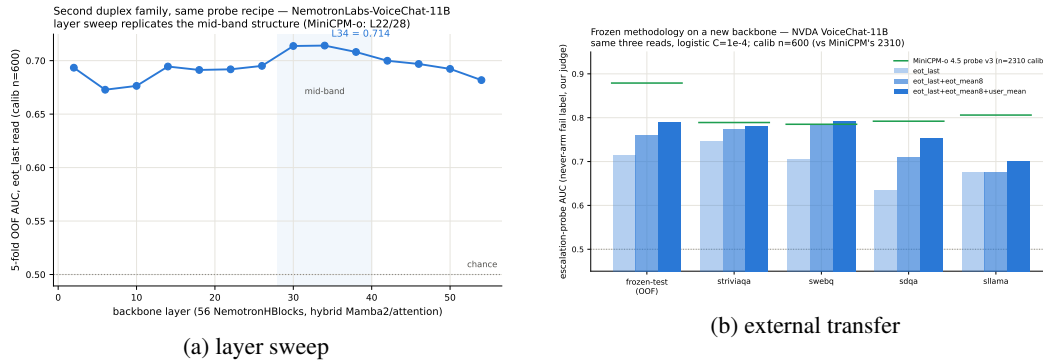


Figure 5: Second-family test: (a) the usable read again peaks mid-network; (b) frozen-recipe transfer lands in the deployed-probe AUC band.

D Query-surface router audit

A RouteLLM-style TF-IDF→LR router under identical labels reaches OOF AUC .669 (below the pool oracle’s .715); a 24-configuration sweep tops out at .689, so this is the query surface’s ceiling, not tuning. Under LOPO it collapses to .38–.57 and scores the 100%-fail trap pool at .230 — it misses the pool a router most needs. The training receipt: at the argmax threshold its out-of-fold accuracy exactly matches the majority class; the internal probes under the same accounting clear majority by 9–29 points (Figure 7). Nor is it data-starved: the identical recipe on RouterBench [Hu et al., 2024] (36,497 prompts) reaches only in-domain AUC .710 with leave-one-benchmark-out at chance; RouteLLM’s released preference-trained routers score .523/.533 zero-shot on our pool and rank the trap pool below ordinary hard pools; our router transfers to RouterBench below chance (.440). On audio labels the router improves (test .814) but still trails the mid-layer probe (.879) and cannot fire mid-stream. In distribution, the gate’s area over random (+.054) is statistically on par with the pool oracle’s (+.042; paired bootstrap delta CI [−.003, +.027], $p=.13$) — the gate’s case is transfer, not the in-distribution margin. $p(\text{True})$ tradeoff variants: Figure 8.

E Gate timing details

Per-layer truncated-forward cost vs. quality (Figure 9): the L22 hidden state is available at ~57–61% of prefill wall-time while L22 is simultaneously the in-mix quality peak; forking much earlier is both weaker and modality-specific, so the fork belongs at ~55–60% depth. If the cloud call launches at

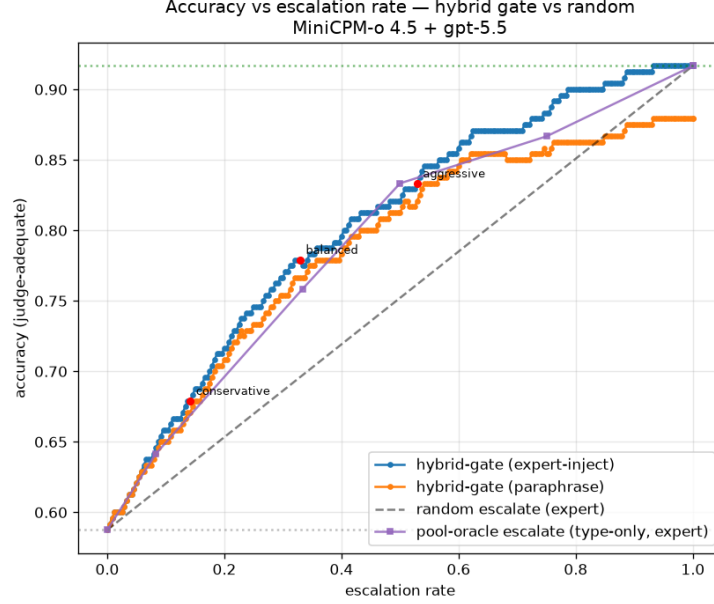


Figure 6: Accuracy vs. escalation rate, frozen test split: gated escalation vs. random-escalation and pool-oracle (type-only) baselines between the small-only and big-only endpoints.

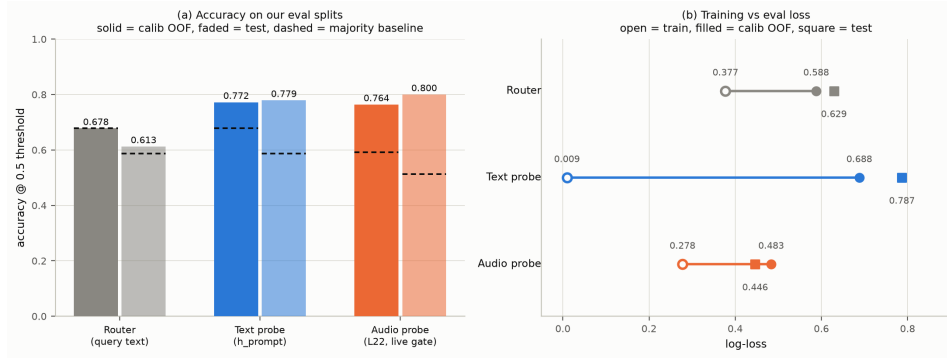


Figure 7: Identical training receipt for the external router and the two internal probes: the router never clears the majority-class baseline; both probes do, by 9–29 points.

502 the gate, the expert result arrives before the local answer finishes for 40–58% of the whole mix but
 503 only ~20–31% of *escalated* queries (they skew to short local answers); the residual wait is P50
 504 2–3 s, P90 ~27 s (Figure 10) — the stall-phrase budget the injection protocol must cover.

505 F Live-loop details

506 **Full live tables.** Table 8 (both scoring views, speakable subset and full pool) and Table 9 (measured response latency). The conservative tier *shortens* the P95 tail: its few escalations displace
 507 exactly the longest local decodes. Figure 11 joins accuracy and latency; Figure 12 replays three real
 508 sessions from recorded timers.
 509

510 **Time to first audible audio.** The sweep above is text-out; a TTS-on re-run of the balanced arm
 511 (same 240 queries, talker head + vocoder on, stall pre-synthesized once in the talker’s own voice)
 512 measures speech onset directly. The gate decision is unchanged — the TTS token enters the context
 513 only after the end-of-turn read (read p50 21 ms / p99 32 ms), and the same 84/240 queries fire. From

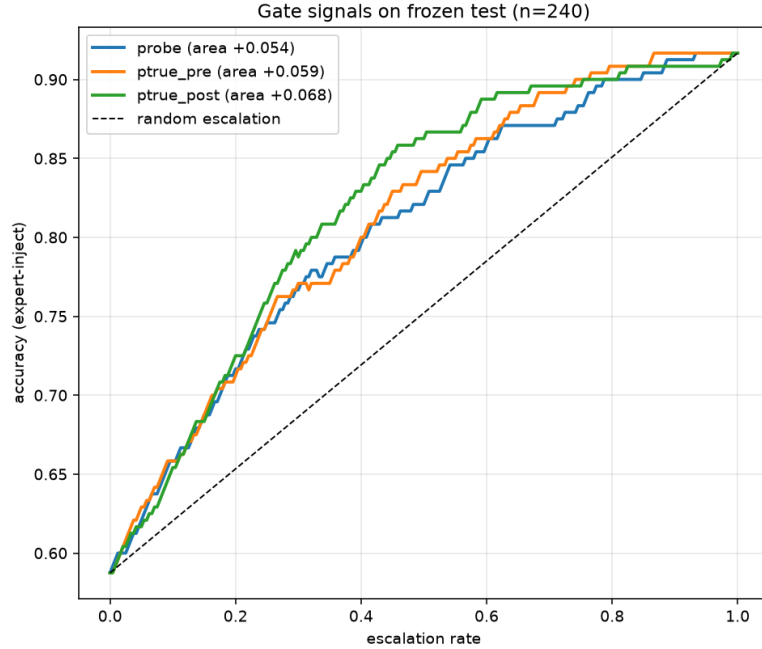


Figure 8: Offline tradeoff for the $p(\text{True})$ signals (pre-answer and post-draft).

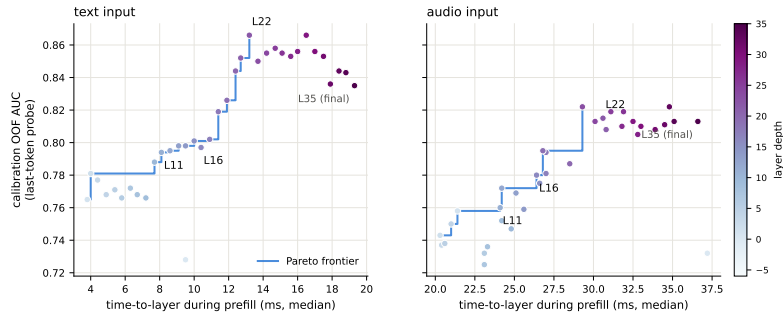


Figure 9: Fork-depth Pareto: per-layer wall-time vs. gate quality.

514 gate decision to first audible audio: **local answers p50 .552 s / p95 .599 / p99 .644; escalated turns**
515 **p50 .022 s / p99 .029** — the canned stall (3.2–4.1 s of speech) plays the moment the gate fires, so the
516 escalated path reaches first audio $25\times$ faster. Expert *content* becomes audible at p50 6.4 s / p95 26.0
517 / p99 34.0 (cache-corrected expert round-trip + first relay PCM, p50 .66 s after the answer returns);
518 the stall covers the head of that wait, and Table 9’s completion profile bounds the tail.

519 **Speculative prefetch, measured and dropped.** Framed as speculative escalation (draft–verify at
520 sentence granularity), its acceptance rate — does an expert answer to the partial transcript answer the
521 full question — is .07/.19/.51 at 25/50/75% of the audio; at the mid-stream gate’s typical fire point
522 6–8 of 10 calls would be wasted. The trap pool’s floor (.00/.10/.27) shares the back-loaded-core
523 property that breaks mid-stream probe reads.

524 **Conflicting-injection controls (172 sessions).** Fabricated conflicting expert answers by within-
525 pool derangement, four framings: worst-case comply is .00 neutral / .53 deployed authority / .79
526 strong-override, while *seeding* words into the talker’s mouth backfires (comply .25, lip-service .62
527 — forced speech is treated as revisable). Math is silently re-derived rather than relayed or disputed.

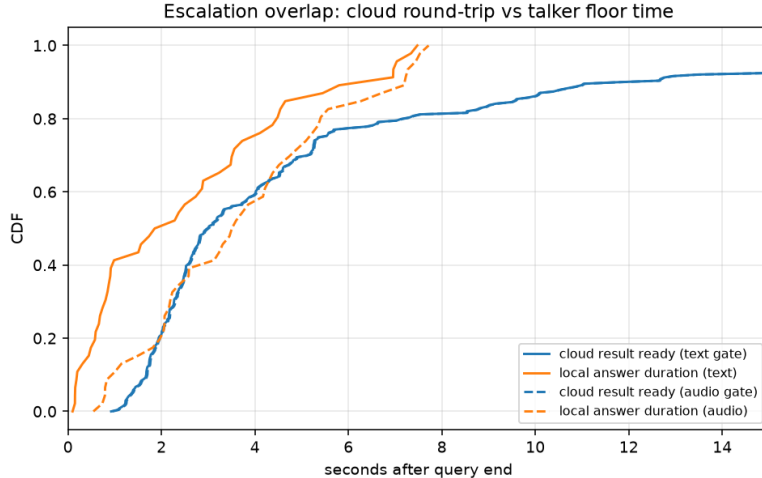


Figure 10: Cloud round-trip vs. local answer duration: probability the expert result arrives before the local answer ends.

Table 8: Live streaming sweep, paired bootstrap 95% CIs ($B=10^4$). Top: speakable subset ($n=218$); bottom: full pool ($n=240$). Gold-inject = escalated rows re-scored with gold-text expert answers; channel cost = paired difference. * CI excludes 0; † CI lower bound at zero.

tier	esc	heard-acc	Δ vs. floor	gold-inject	channel cost
never	0%	.422 [.36,.49]	—	.422	—
conservative	14%	.528 [.46,.59]	+106*	.550	+.023†
balanced	31%	.592 [.53,.66]	+170*	.661	+.069*
aggressive	57%	.633 [.57,.70]	+211*	.761	+.128*
always (synth)	100%	—	—	.922 [.89,.95]	—
never (full)	0%	.383 [.32,.45]	—	.383	—
conservative	16%	.483 [.42,.55]	+100*	.525	+.042*
balanced	35%	.554 [.49,.62]	+171*	.654	+.100*
aggressive	61%	.596 [.53,.66]	+212*	.771	+.175*
always (synth)	100%	—	—	.917 [.88,.95]	—

Model-wrong $\times$ wrong-injection swallow rate is .64: expert quality is the accuracy ceiling of the cascade.

Uplink diagnostics. On the escalated population the expert answering the talker’s own transcript scores .585 vs. .871 on gold text. Prompt-level rescues on the self-transcript are no-ops (the damaging errors are internally-coherent entity substitutions); replacing the ears helps: open-weights Whisper recovers little (+4 pp) but a stronger cloud ASR on the same audio recovers 38% of the gap (.585 $\rightarrow$.694, McNemar $p=.007$) — diagnostic only (its critical-path latency and external-pool survival are unmeasured). A same-model gold-text control with an audio-capable expert shows the audio channel itself is lossless for speakable content: the leak is the talker’s transcription, not the synthesis or the relay (relay drops: 5 cases of 108; corrupted transcripts dominate the escalated heard-failures, first-generation sweep).

Boundary query classes. *False premises:* the model fails substantially (.63 text / .47 audio) and the expert would help (.80), but the gate is blind (fire@balanced .18 vs. trap’s .90; no score separation) — answering along a false premise does not feel like inability. *Real-time queries* (“what is NVDA trading at”): the gate is not blind, the class was simply absent from every static training family; extending calibration with FreshQA [Vu et al., 2023] (fast-changing labeled escalate a priori, never-changing as in-family controls) raises held-out fast-changing fire rates to .80/.93/1.0 across

Table 9: Measured response latency (s), $n=240$ per arm (latency is channel-independent). P99 on $n=240$ is indicative only.

tier	mean	P50	P95	P99	Δ mean	Δ P50	Δ P95	Δ P99
never	4.6	2.0	15.8	17.8	—	—	—	—
conservative	4.6	2.6	12.8	23.0	+0.0	+0.6	-3.0	+5.2
balanced	5.8	3.5	15.9	32.7	+1.2	+1.5	+0.1	+14.9
aggressive	6.6	4.4	18.5	33.0	+2.0	+2.4	+2.7	+15.2

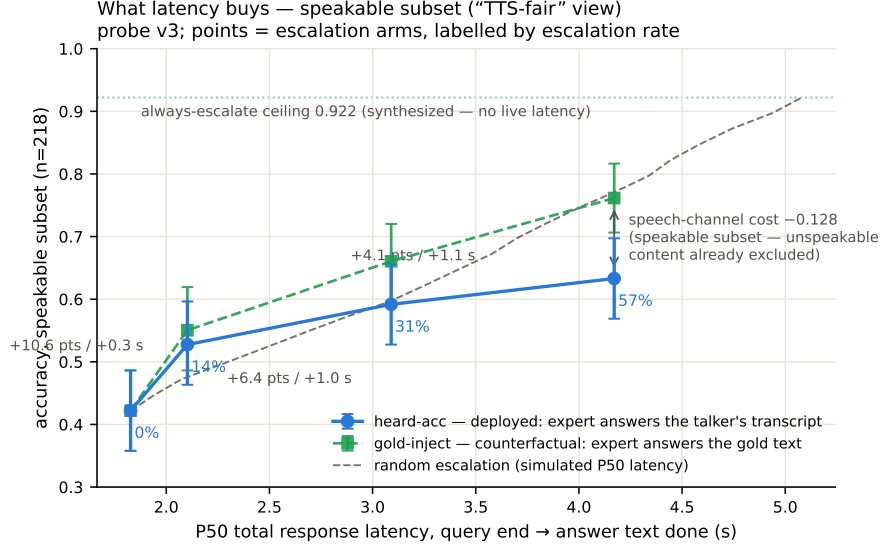


Figure 11: Accuracy vs. per-arm P50 latency (speakable subset), both scoring views, with the simulated-P50 random reference.

tiers with frozen-test AUC unchanged (.877 vs. .879). One calibration subtlety: tier budgets must remain quantiles of the deployment-mix scores, or the new capability is spent on its own threshold shift.

Demonstration system. The pipeline runs as an interactive web demo on one H100: live microphone audio, end-of-turn read (~54 ms) against the balanced threshold, escalated turns stall, consult the expert, and *speak* the relayed answer through the model’s native TTS head (probe context and thresholds untouched). Two demo-only extensions: a web-search tool for the expert (real-time queries then return live values), and backchannel-aware barge-in — speech over the talker ducks playback, a fast ASR pass returns the floor on backchannels (“ok”, “mm-hm”; an English+Chinese lexicon) or commits the interrupt, aborting generation at chunk granularity and seeding the next turn. All measured arms keep the tool-free expert and turn-based loop; the demo will be made available upon publication.

G Duplex-validation sweep: the gate under the talker

The headless live sweep (§6) never engages the talker head; this sweep replays its full 240-query pool through the deployed voice loop above to test whether the frozen probe read survives the spoken regime. Two arms, one session per query, deployed v4 gate with scores logged but escalation disabled (the arm isolates gate validity, not routing outcomes). **Clean:** the query streams as mic-shaped PCM (pool TTS audio plus steady noise at RMS .008), the energy VAD ends the turn, the talker answers aloud, and the session is cut after the first audible chunk. **Overlap:** an easy warmup question (probe score <.25, locally answered, >5 s spoken answer) puts the talker mid-answer, and

Three live sessions, timestamped (balanced arm; every event time is a recorded live-loop timer)

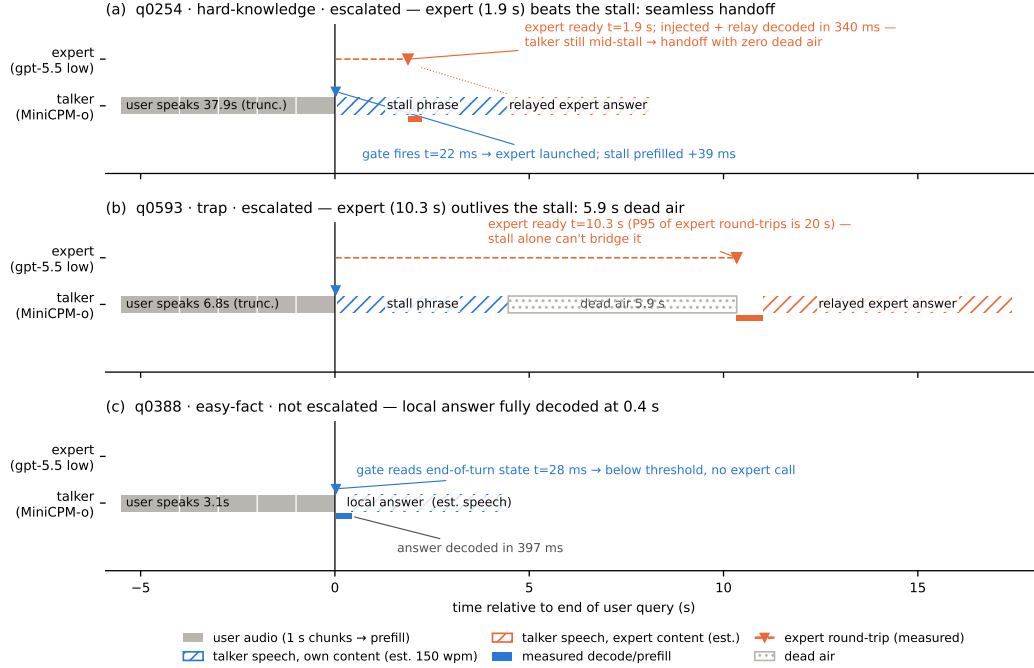


Figure 12: Three real sessions from recorded timers: expert beats the stall phrase (seamless); expert outlives it (dead air, the P99 signature); gate stays closed (local answer fully decoded 0.4 s after end of turn). Speech occupancy is estimated at 150 wpm (hatched); the measured arms are text-out, so no vocoder onset is plotted.

the target query is then spoken over it; sustained overlapping speech commits a barge-in whose audio seeds the next turn. The barge-in path engaged on 237/239 pairs.

Against the frozen local-failure labels of the same pool, the identical frozen probe reads: headless AUC .866 [.817, .909], clean .871 [.824, .914], overlap .856 [.803, .901]; barged-only subset ($n=237$) .854. Paired bootstrap deltas vs. headless are $+.005$ [$-.025, +.035$] (clean) and $-.009$ [$-.036, +.016$] (overlap) — both inside the $\pm .02$ –.03 live replication floor. Per-query scores correlate across regimes at $r=.90$ –.92, and the balanced threshold makes the same fire/hold decision as the headless read on .887 (clean) and .912 (overlap) of queries. The end-of-turn read costs 24–26 ms at p50 ($p95 \leq 30$ ms) in both arms, with the first audible answer chunk arriving ≥ 0.5 s after the gate decision ($p50 \approx 0.58$ s), so the decision precedes any audible commitment. Scope: the loop keeps per-turn session control — the mic stays open while the talker speaks and interruptions seed the next read, but incoming audio is not prefilled *during* generation; validating a probe read concurrent with generation remains open.

H Fixed-threshold transfer and the expert-benefit oracle

Both analyses below run in the *gold-inject* view (escalated rows take the expert's answer to the gold text). The deployed heard view cannot evaluate a freely moving threshold: the most permissive arm escalates 50%, so the other half of each pool has no expert outcome on record, whereas the ceiling runs cover every id. Labels follow each pool's own judge (Table 3); the two judges disagree by up to 13 points on Web Questions, so they are never mixed within a row. AUC columns are recomputed from the deployed sweep's stored end-of-turn scores; they differ from Table 3's AUC column — captured in a separate scoring pass — by up to .03 externally and .04 on the internal pool (.840 here vs. .879 there): the passes render and prefill the audio independently, and the probe score carries

render-to-render jitter. AlpacaEval is excluded — it has no binary correctness label on either side, only a 1–5 score whose expert ceiling is degenerate (195/199 at 5).

One absolute threshold, six pools. The deployed gate adapts per pool, but only through label-free quantiles of its own score distribution. To separate what the probe’s *ranking* carries from what its *scale* carries, we freeze the single absolute threshold the internal system actually deployed ($\tau=.628$, the balanced tier) and apply that number verbatim everywhere. **Scale does not transfer:** the same τ realizes escalation rates from 2.0% (Reasoning QA) to 48.0% (SD-QA), a 24-fold spread, so a fixed cut cannot hold an escalation budget across pools. **Ranking does:** at whatever rate it fires, fixed- τ escalation beats a random reference at the same realized rate on four of five external pools, paired-bootstrap interval excluding zero (Table 10). The exception is Reasoning QA, where τ fires on 2% of queries and there is almost nothing to measure. This is the concrete case for the label-free quantile step: it buys budget control, not discrimination.

Does the gate find failures the expert can fix? The probe is fit and read against *local failure*, but what a routing system needs is *expert benefit*: $y=1$ when the local model is wrong *and* the expert is right. Re-scoring the same frozen scores against this harder label costs little on the retrieval pools (Speech TriviaQA .796→.782, Llama Questions .830→.813) and more where the expert is itself weak (Web Questions .773→.689 against a .864 expert ceiling; internal .840→.732). The gate thus remains a usable benefit predictor out of distribution (.63–.81) despite never having seen an expert outcome, and the drop is largest exactly where the paper already reports a flat live curve. On Web Questions the mechanism is direct: 75.0% of local failures are expert-fixable pool-wide, but only 65.5% of the failures inside the gate’s top-scoring 15% are, and that share climbs monotonically as the cut widens (.655/.685/.711 at 15/30/50%) — the gate’s most confident failure calls are the ones the expert also misses. Table 11 places the deployed gate against the matched-budget oracle that escalates true benefit cases first: the gate clears random at every pool and budget, capturing a fifth to three fifths of the random-to-oracle span, and the residual gap is the headroom a decode-time or expert-aware second stage would have to claim.

Table 10: One absolute threshold ($\tau=.628$, frozen on the internal pool) applied verbatim to every pool; gold-inject view. Δ is fixed- τ accuracy minus a random reference at the *same realized rate* (paired bootstrap over ids, $B=10^4$, seed 42). AUC columns re-score the same frozen scores against local failure and against expert benefit ($y=\text{local wrong} \wedge \text{expert right}$, base rate in the last column).

pool	n	rate	local	acc	rand	Δ 95% CI	AUC _{fail}	AUC _{benefit}	base
Speech TriviaQA	250	.284	.664	.832	.750	[+.044, +.121]	.796	.782	.308
Speech Web Questions	250	.360	.568	.720	.675	[+.008, +.084]	.773	.689	.324
Llama Questions	250	.080	.836	.868	.843	[+.006, +.046]	.830	.813	.112
SD-QA	200	.480	.515	.805	.714	[+.045, +.136]	.800	.765	.435
Reasoning QA (zh)	202	.020	.589	.609	.595	[−.001, +.035]	.679	.634	.302
internal frozen test	240	.350	.383	.654	.570	[+.038, +.131]	.840	.732	.533

Table 11: Oracle frontier at matched escalation budgets (gold-inject view): random reference / deployed quantile gate / benefit oracle, which escalates the true $y=1$ cases first. The gate clears random everywhere; the oracle column is the remaining headroom.

pool	15% budget			30% budget			50% budget		
	rand	gate	oracle	rand	gate	oracle	rand	gate	oracle
Speech TriviaQA	.710	.760	.816	.755	.836	.964	.816	.912	.972
Speech Web Questions	.612	.640	.720	.657	.700	.868	.716	.764	.892
Llama Questions	.850	.888	.948	.864	.912	.948	.882	.932	.948
SD-QA	.577	.635	.665	.639	.745	.815	.723	.820	.950
Reasoning QA (zh)	.631	.678	.738	.674	.728	.891	.730	.767	.891
internal frozen test	.463	.492	.533	.543	.608	.683	.650	.742	.883

I Transfer latency shapes

Matched-rate precision. Precision at a fixed escalation rate is capped at base-fail/rate: on Llama Questions (base fail .16) the 50% cap is .32 and the deployed probe reaches .304 — 95% of the cap (recall .93). Quoting raw precision would misread the strongest pool as the weakest.

The median left-fold. On easy pools per-arm P50 is non-monotonic (Llama Questions 1.52→1.17→1.60/1.42 s): a verified median-of-a-mixture mechanism, not noise. The probe preferentially escalates the queries whose *local* decode is longest — on retrieval pools wrong answers hedge at length (decode time and answer length $r=.97$), so a wrongness-selecting probe strips the slow tail as a side effect (escalated-at-15% ids: never-arm local P50 1.711 s vs. 1.198 s kept; local accuracy .32 vs. .73). Controls: the probe does not key on length ($\text{corr}(\text{score}, \text{length}) = +.05$; ≈ 0 conditioned on wrongness), and on SD-QA — where wrong answers are short — the same probe selects wrongness identically and produces no fold. The mean is monotonic (1.65→2.06 s); kept-local P50 falls 1.52→0.43 s across tiers (Figure 13). On Reasoning QA the shape inverts usefully: the local chain-of-thought is *spoken*, so every reasoning token enters the response clock, while the expert’s chain is hidden behind a flat ~ 3.7 s round-trip — escalation truncates the latency tail (P90 13.25→10.94 s) rather than fattening it.

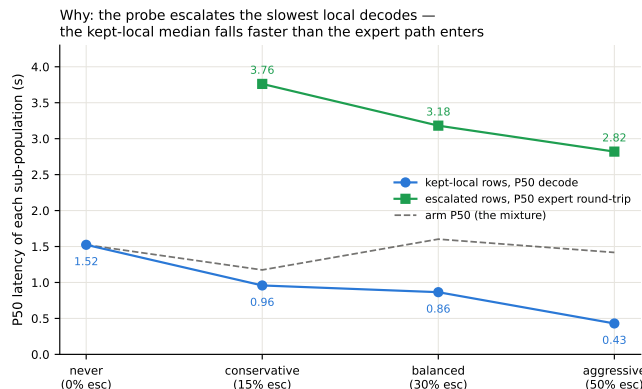


Figure 13: Mixture decomposition of Llama Questions latency per arm: kept-local P50 falls 1.52→0.43 s as the probe strips the slowest local decodes while escalated rows pay a flat ~ 3 s expert round-trip; the arm median (dashed) is their mixture.

AlpacaEval scale caveats. The official 4.8 is an offline chat-mode number — our chat-mode control reproduces it at 4.86 — and the chat-vs-live gap (4.86 vs. 3.99) is the duplex loop’s spoken-reply style plus a token cap, so only live arms are comparable to each other. Selected queries score 3.90 on the never arm vs. 3.98 for skipped: no discrimination; escalated rows gain (3.90→4.71) because the expert writes better long-form answers — random picks would harvest the same gain.

J Full-pool and per-family figures

Figures 14 (dual-view) on the full frozen pool, and the complete external-family figure set (Appendix K); the noise audit behind the $\pm .02$ –.03 floor is Figure 15.

K External-benchmark figure set

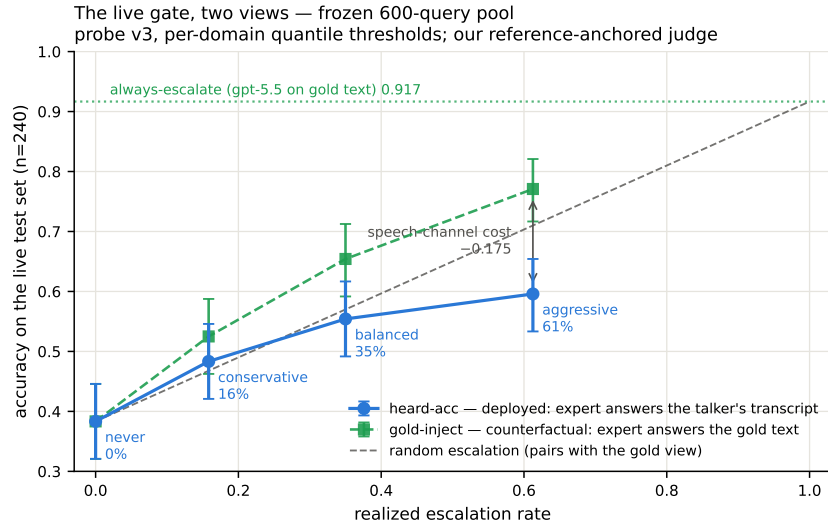


Figure 14: Dual-view live tradeoff, full pool ($n=240$); companion to Figure 3.

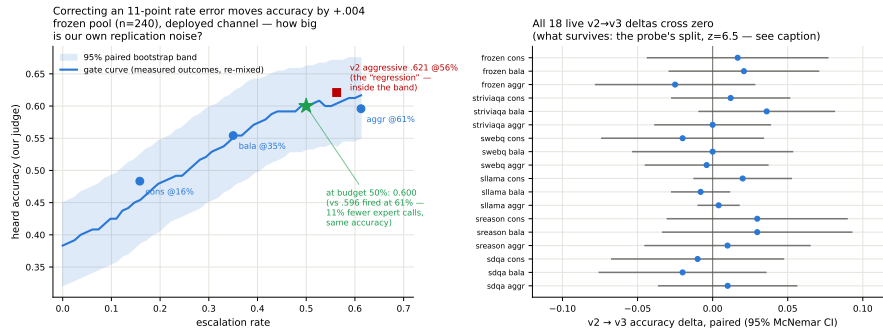


Figure 15: Replication-noise audit: same query, same audio, re-run across arms. Kept-local verdicts flip 2.3–18.8% by pool; repeated escalation 0.7–10.6%; implied paired SE .009–.028.

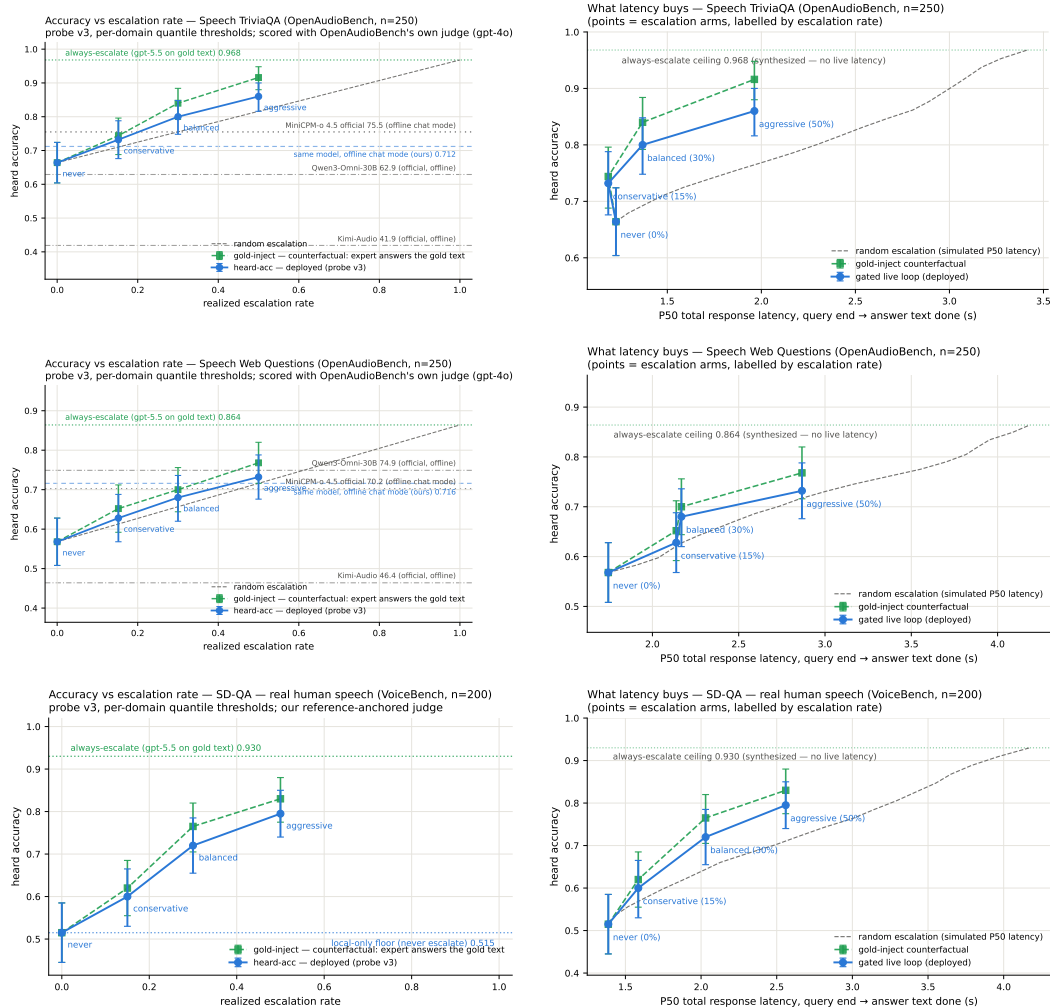


Figure 16: Speech TriviaQA, Speech Web Questions, SD-QA: dual-view (left) and accuracy–latency (right).

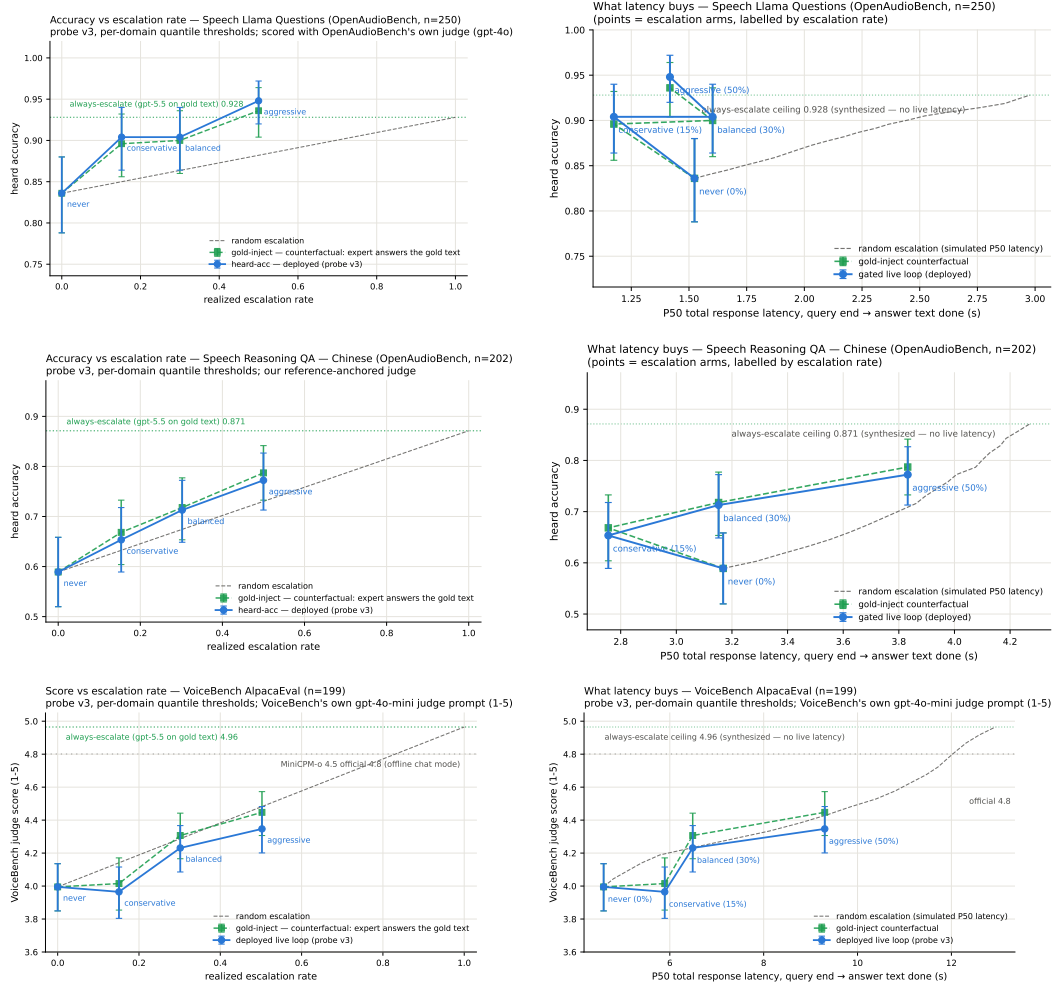


Figure 17: Llama Questions, Reasoning QA (zh), VoiceBench AlpacaEval: dual-view (left) and accuracy-latency (right).

638 L Judge and self-evaluation prompts

639 All adequacy labels use the following fixed system prompt (structured outputs with a strict {reason,
640 adequate} schema; the judge never sees which system produced the answer):

```
641     You are a strict, fair grader of a small language model's answers.  
642     You are given a QUESTION, an optional REFERENCE answer, and a  
643     CANDIDATE answer produced by a small model. Decide whether the  
644     CANDIDATE is ADEQUATE, meaning ALL of: (1) factually correct;  
645     (2) any reasoning shown is valid; (3) it actually and completely  
646     answers the question that was asked. For multiple-choice questions  
647     the candidate must land on the correct option (by letter or by  
648     unambiguously stating the option's content). A partially correct,  
649     hedged, refused, off-topic, or truncated answer is NOT adequate. If  
650     a REFERENCE is provided, grade against it. Be strict but do not  
651     penalize harmless differences in format or extra correct detail.
```

652 The verbalized self-evaluation baselines use one fixed phrasing each; $P(\text{Yes})$ is read from the first
653 generated token's distribution:

```
654     [pre] I am going to show you a question. Do NOT answer it. Judge  
655     honestly whether you yourself would answer it correctly. Question:  
656     {q} Would you answer this question correctly? Reply with exactly one  
657     word: Yes or No.  
658     [post] Here is a question and a proposed answer. Question: {q}  
659     Proposed answer: {a} Is the proposed answer correct? Reply with  
660     exactly one word: Yes or No.
```

661 M Reproducibility

662 All sampling and splits use a fixed seed (42); the calibration/test split is frozen once and the test split
663 is touched only by the evaluation scripts. Environment pins: torch 2.8.0; transformers 4.51.0 for
664 MiniCPM-o 4.5, with per-model pins for comparison backbones; all experiments run on single-H100
665 instances. Probe artifacts (weights, feature configuration, per-pool label-free thresholds), the judge
666 prompt above, all trace files, and the per-experiment execution log — including per-run GPU and
667 API spend, on the order of \$500 total — ship in the project repository, released upon publication.

TODO (process only — delete before submission)

- [TODO: P1 — DONE 2026-08-25 (external review applied): main text now ends within 8 pages (refs start mid-page 8). Restructure: 3 contributions + hero figure; merged signal section; live 6pp → 1.2pp; router audit 3 lines; title dropped “Zero-Training”; causal claims softened (“consistent with”); no version numbers / artifact filenames in main text; everything cut lives in the appendix (A–L) or git history]
- [TODO: P2 — peer + meeting feedback (GPT reviewer pass 2026-08-26, borderline 4/7). (a) and (b) DONE 2026-08-26; (c) dropped as not computable; (d) unchanged. (a) DONE — three claims narrowed: “routing beats scaling the speech backbone” → “selective cloud escalation outperforms larger standalone backbones”; “standard readouts fail” → “late-layer readouts turn brittle after duplex fine-tuning” (abstract, intro contribution 1, hero + teaser panel-1 titles); and the EOT loop is no longer equated with native full-duplex — intro now carries an explicit scope sentence (“*duplex* names the training regime, not the interaction protocol”), live.tex states the turn-based protocol up front and is retitled “Live End-of-Turn Escalation”, and “duplex session/system” → “spoken session”/“live system” throughout. (b) DONE — new appendix section (app:fixedthr) + two tables: one absolute threshold ($\tau=.628$, frozen internally) applied verbatim to all pools fires at 2–48% yet still beats matched-rate random on 4/5 (ranking transfers, scale does not); expert-benefit oracle ($y = \text{local-wrong AND expert-right}$) with AUC re-scored per pool and a gate/random/oracle frontier at 15/30/50% budgets. Script: scripts/09_fixed_threshold_oracle.py → figures/fixed_threshold_oracle.json. Gold-inject view (heard view only observes the ~50% of ids some arm escalated); labels follow each pool’s own judge (oab_ok vs adequate — they differ by 13pp on Web Questions, never mix); AlpacaEval excluded, it has no binary label on either side. (c) DONE 2026-08-26 (was DROPPED, then re-run with TTS on): bench_live gained a tts flag (init_tts + Token2wav, canned stall, gen_speak port of demo_app); balanced arm re-run on the frozen 240 (~50 min H100, expert cache hits). EOT→first-audible: local p50 .552s/p99 .644s, escalated (canned stall onset) p50 .022s/p99 .029s, expert-content p50 6.4s/p95 26.0/p99 34.0. Same 84/240 fire as text-out; probe read unperturbed (p50 21ms). Landed: live.tex sentence + app:livedetail paragraph. Traces data/frozen_v3tts_traces.jsonl.balanced.shard0-3; analysis scripts/10_tts_latency.py. Fixed en route: fig:timeline-live panel (c) was captioned “first token 0.4s” but the plotted value is answer_ms = *full* answer decode; caption + figure title corrected and the figure regenerated. (d) PARTIAL DONE 2026-08-26 (was SKIP): duplex-validation sweep ran — the frozen-pool 240 replayed through the DEPLOYED voice loop (talker on, spoken answers) in clean + overlap/barge-in arms via _ws_duplex.py against the live demo; frozen v4 probe, no refit: AUC .871/.856 vs .866 headless (paired deltas inside the replication floor), decision agreement .89/.91, barged 237/239. Landed: live.tex closing paragraph, app:duplexval, intro scope sentence updated, Limitations (vi) narrowed to “concurrent-with-generation read remains open”. Data: gate-data volume duplex_sweep/{clean,overlap}.jsonl; analysis _duplex_analyze.py. This also partially recovers (c): the voice loop logs EOT→first-audible per turn (first_audio_client_ms, p50 ≈.58s, both arms) — usable if a reviewer asks, though only under probe-off replay, not the escalating sweep. Still skipped: human continuity study, second expert, model-level concurrent listen/speak read. Frame the paper as what it is: pre-response competence probing in duplex-trained models + turn-based live escalation, now with a deployed-loop validity check.]
- [TODO: P2.2 — meeting feedback round 2 (2026-08-26), four items, all landed same day: (1) ugly zigzag figures: fork_pareto redrawn as depth-colored scatter + Pareto frontier (figures/fork_pareto.py, new); sllama_latency_decomp cut to the clean single decomposition panel (median-vs-mean fold fact stays in prose). (2) main table:

720 **tab:transfer upgraded Kitty-style — live tiers + ceiling + AUC + official PLUS gold-**
721 **inject@30% rand/gate/oracle block (numbers from tab:oracle); judge folded into row**
722 **names. Views kept separate and labeled — do NOT let anyone compare live heard**
723 **vs gold-inject across blocks. (3) special-token routing discussed in related work:**
724 **pause/thought tokens (goyal2024pause, zelikman2024quietstar), fused think modes**
725 **(yang2025qwen3, guo2025deepseekr1), MiniCPM-o’s own think gate — they decide**
726 **more compute not which model, need talker fine-tuning, policy baked per checkpoint;**
727 **ours = frozen + pluggable + beats the built-in gate at matched budget (tab:policy).**
728 **(4) simple-but-important framing: intro minimalism sentence (“competing signals**
729 **read the wrong place / arrive too late / must be trained in”). Page budget paid by:**
730 **fair_dualview .56→.45 linewidth, caption + related + gap compressions.]**

- 731 • **[TODO: P3 — submission day (8-28): delete this section; final anonymization grep**
732 **(names, rhe9527, modal URLs, gmail); decide public-repo → private; OpenReview**
733 **form + upload (deadline 8-29 AoE, moderation hard limit Fri 8-28)]**